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**MID TERM EXAMINATION**

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

1<sup>st</sup> Semester, September - October 2025**Paper Code: BS-105****Subject: Applied Physics-I****Time: 1½ Hrs.****Max. Marks: 30**

**Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining. Assume the data if required.**

| Q. No.            | Question                                                                                                                                                                                                                                            | Max. Marks | CO(s)    |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| <b>Question 1</b> |                                                                                                                                                                                                                                                     |            |          |
| <b>1 (a)</b>      | Give Kelvin-Planck and Clausius statements for second law of thermodynamics.                                                                                                                                                                        | <b>2.5</b> | <b>1</b> |
| <b>1 (b)</b>      | Calculate the entropy change when 1 g of ice at $-10^{\circ}\text{C}$ converts into water at $0^{\circ}\text{C}$ . Given: Specific heat of ice = 0.5, latent heat of ice = 80 Cal/g.                                                                | <b>2.5</b> | <b>1</b> |
| <b>1 (c)</b>      | Check if the vector field $\mathbf{A} = -(2z^2\mathbf{y} / x^3) \hat{i} + (z^2 / x^2) \hat{j} + (2yz / x^2) \hat{k}$ is irrotational or not.                                                                                                        | <b>2.5</b> | <b>2</b> |
| <b>1 (d)</b>      | In what way the displacement current is different from the conduction current? Explain.                                                                                                                                                             | <b>2.5</b> | <b>2</b> |
| <b>Question 2</b> |                                                                                                                                                                                                                                                     |            |          |
| <b>2 (a)</b>      | Using First law of Thermodynamics, derive an expression for adiabatic equation of state $PV^{\gamma} = K$ for a perfect gas, where $K$ is a constant.                                                                                               | <b>5</b>   | <b>1</b> |
| <b>2 (b)</b>      | 1 mole of Dry air at $32^{\circ}\text{C}$ is compressed (i) slowly and (ii) suddenly to $\frac{1}{4}$ of its volume. Find the change in temperature and work done in each case. Given $\gamma = 1.4$ for dry air and $R = 8.3 \text{ J/deg mole}$ . | <b>3</b>   | <b>1</b> |
| <b>2 (c)</b>      | State and prove Carnot's theorem for a reversible heat engine.                                                                                                                                                                                      | <b>2</b>   | <b>1</b> |
| <b>Question 3</b> |                                                                                                                                                                                                                                                     |            |          |
| <b>3 (a)</b>      | State and prove Stokes' theorem in vector calculus.                                                                                                                                                                                                 | <b>5</b>   | <b>2</b> |
| <b>3 (b)</b>      | Derive Maxwell's equations in differential and integral form for vacuum / free space.                                                                                                                                                               | <b>5</b>   | <b>2</b> |
| <b>Question 4</b> |                                                                                                                                                                                                                                                     |            |          |
| <b>4 (a)</b>      | Show that the energy transferred by a wave per unit area per unit time is given by $I = \frac{1}{2} \rho v \omega^2 A^2$ .                                                                                                                          | <b>5</b>   | <b>2</b> |
| <b>4 (b)</b>      | A Carnot engine has an efficiency of 40% when its sink is at $27^{\circ}\text{C}$ . What is the temperature of the source? By how much should the source temperature be raised to increase efficiency to 50%?                                       | <b>5</b>   | <b>1</b> |

\*\*\*\* End of Question Paper \*\*\*\*